

Government Dyal Singh Graduate College

University of Punjab, Lahore



Final Year Project Documentation

REVOLUTIONIZING DIGITAL LEARNING

WITH AI

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REVOLUTIONIZING DIGITAL LEARNING WITH AI

Team ID: FYP-09

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STATEMENT OF SUBMISSION

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ABSTRACT

Existing digital Learning platforms like Udemy and Coursera are designed for general education and lack customization for specific departments such as IT. Students rely on scattered online resources and physical notes, making learning inefficient, while teachers face challenges in organizing programming courses, and uploading lectures. With limited IT course resources available in Urdu, there is a clear need for a department-focused platform. This project presents “Revolutionizing Digital Learning With AI”, a web-based E-Learning System developed specifically for BS.IT Students, and Teachers of the University of the Punjab. The system supports three stakeholders: Students, Teachers, and Admin. Key objectives include user authentication with email and Google via Firebase, profile management, course and lecture delivery, assignment sections, an AI Chatbot for queries, and dedicated dashboards for students, teachers, and admin. The platform enables students to view video lectures and content aligned with University of the Punjab BS.IT degree program along with assignments. Teachers can create courses, lectures, and assignments, while admins manage courses and users. The scope is limited to web development for FYP. The developed Revolutionizing Digital Learning With AI system provides a centralized, secure, and department-specific solution that improves learning efficiency for IT students and simplifies content delivery for teachers. It bridges gap between global platforms and the localized academic needs of IT departments.

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CHAPTER 1

INTRODUCTION

1.1 Background

Our Project belongs to the field of Education Technology and Revolutionizing Digital Learning (RDL), which is used to manage, deliver, and monitor education content digitally.

Many existing Revolutionizing Digital Learning platforms such as Udemy and Coursera, are designed for general educational purpose or use and are not customized for specific departments, such as the IT Department, creating difficulties in organizing programming courses, lecture materials, coding resources, and semester wise content.

In traditional systems, students rely on physical notes and scattered online resources, making learning inefficient, while teachers face challenges in creating courses, uploading lectures, and tracking student progress.

1.2 Problem Statement

Because of the emerging trends of e-learning and low to no availability of resources for IT courses in the local language, Urdu. Our team decided to create a platform for IT Professionals, whether students, teachers or skilled learning person. This platform will help students to view video lectures along with content of Punjab University's IT degree Program, along with assignments. This will help both students to learn and teachers to teach.

1.3 Objectives

- User Authentication with email and Google (implement using firebase).
- Profile section to view personal details.
- Course and Lectures for the user to learn.
- Assignment section for the user to test and retain their knowledge.
- AI Chatbot page to respond to user queries.

- A teacher page or Dashboard to create courses, lectures and assignments.
- An admin page or Dashboard to manage courses and users.

1.4 Scope of the Project

For FYP, the scope is limited to Web development. No Android version. In future they'll be included.

- The following will be stakeholders:
 - Students
 - Teachers
 - Admin
- For the online database, MongoDB will be used to manage users, courses, lectures and assignments.
- YouTube will be the source of video lectures
- For development, VS Code with JavaScript, HTML for formatting lecture content, and CSS for styling.
- Separate pages or Dashboard for teacher and admin.

Table 1.1 Tools and Technologies Used

Category	Tools & Technology
Programming Language	JavaScript
Web Frameworks	React, Express (Node.js)
Database	MongoDB
AI	Open AI (Gemini API)
Hardware / IOT Platform	Laptop, Internet Connection
Cybersecurity	JWT Authentication, Password Hashing (bcrypt)
Other tools	Git / Git Hub, Postman (API Testing)

CHAPTER 2

LITERATURE REVIEW / RELATED WORK

2.1 Summary of similar existing systems or research

P. Shukla and Y. Pandey [1] reviewed the role of Learning Management Systems (LMS) in higher education and highlighted their importance in supporting e-learning and blended learning. They explained how LMS platforms enhanced teaching, learning, communication, and student engagement through digital tools and resources. Despite these benefits, many institutions faced challenges such as poor internet connectivity, lack of awareness, insufficient training, and resistance to technology. Additional barriers included high implementation costs, limited funding, inadequate learning materials (especially in Moodle), and traditional paper-based teaching cultures. The study concluded that addressing these challenges through training, infrastructure development, and institutional support was essential for effective LMS adoption in higher education.

Djafri, N., Sarkawi, S., & Rusiadi [2] reviewed the application of Learning Management Systems (LMS) in distance education and highlighted their growing importance in the digital era. They explained that LMS enhanced learning flexibility, quality, and interaction through features like material management, assessments, and online communication. The study also identified major challenges in LMS implementation, including limited internet access, low technological literacy, and cost constraints. Findings showed that LMS was especially crucial during the COVID-19 pandemic to ensure continuity of education. Overall, the authors concluded that the success of LMS depended on strong policy support, adequate infrastructure, and proper training for educators and learners.

Marpelina, L., Sariyatun, & Andayani [3] reviewed how digital technology was transforming history learning in the era of Industry 4.0 by shifting from

traditional memorization to interactive, student-centered approaches. They highlighted the use of digital tools such as e-books, virtual reality, augmented reality, social media, YouTube, and podcasts to create engaging learning experiences. The study emphasized a “boundless time and place” learning environment that allowed students to learn history anytime and anywhere. It also stressed the crucial role of history teachers in developing students’ digital literacy, critical thinking, and ethical use of online resources. Overall, while digital history education offered greater access and engagement, it also faced challenges such as unequal technology access and the need for continuous teacher training.

Rofiuddin, & Aulia, F. [4] reviewed an article that presented a systematic literature review on Hybrid Learning Management Systems (HLMS), focusing on trends, benefits, and existing research gaps. They explained that HLMS combined face-to-face and online learning to enhance flexibility, learner engagement, and instructional effectiveness. The review identified key advantages such as personalized learning, improved accessibility, and better use of digital technologies in education. However, it also highlighted research gaps related to system design, evaluation models, user experience, and long-term effectiveness. Overall, the study emphasized the need for further empirical research to optimize hybrid LMS implementation in diverse educational contexts.

Bitar, Nizar, and Nitza Davidovich [5] reviewed how Digital Learning transformed higher education in other countries, accelerated by the COVID-19 pandemic. Using in-depth interviews with 15 lecturers, they identified four key domains shaping implementation: educational, personal, cultural/social, and institutional. Digital Learning enhanced pedagogical innovation, accessibility, and student engagement but required major curriculum redesign and new teaching roles. Lecturers faced challenges such as increased workload, technological dependence, emotional strain, and cultural adaptation. The study recommended culturally responsive digital strategies, integration of traditional practices like chavruta, and stronger institutional support.

Amadoki, Daniel [6] reviewed students' perspectives on factors influencing the use of e-Learning Management Systems (e-LMSs) in public universities in Ghana. Using a cross-sectional survey of 531 students, the study examined teaching activities, administrative activities, system effectiveness, learning technologies, and perceived behavioral control. Findings showed that students generally had positive perceptions of all these contextual factors and felt confident using e-LMSs. Students reported that e-LMSs effectively supported teaching, assessments, communication, and administrative functions. The study recommended continuous student training, improved technological infrastructure, and full integration of e-LMSs into university teaching and administration.

Adzharuddin, N. A., & Ling [7] reviewed the effectiveness and acceptance of Learning Management Systems (LMS) among university students. They explained how LMS supported teaching and learning by enabling content delivery, communication, assessment, and learning management beyond the traditional classroom. Prior studies showed that LMS could enhance accessibility, interaction, and learning efficiency, but its impact depended on how instructors and students used it. Challenges included limited use of interactive features, lack of technical skills, infrastructure constraints, and insufficient training. The study concluded that LMS was essential in modern universities and recommended proper training, institutional support, and effective implementation to maximize its benefits.

Verhoef, P. C., Broekhuizen [8] reviewed the role of digital and online learning technologies in transforming higher education teaching and learning practices. They reviewed how universities adopted e-learning and blended learning models, particularly in response to recent global disruptions. The study highlighted benefits such as increased learning flexibility, continuity of education, and improved access to learning resources. It also identified challenges including digital inequality, limited technological readiness, and increased demands on instructors and students. The authors concluded that

sustainable digital education required strategic planning, investment in infrastructure, and ongoing pedagogical support.

Ray, Suvendu, and Deb Prasad Sikdar [9] reviewed how Artificial Intelligence (AI) could be integrated into the flipped classroom model to enhance digital pedagogy and personalized learning. They reviewed existing studies showing that AI tools such as intelligent tutoring systems, virtual tutors, and adaptive platforms improved student engagement and learning outcomes. The study highlighted benefits including personalized learning paths, timely feedback, effective use of class time, and increased accessibility and scalability. Key challenges identified included technological infrastructure limitations, teacher preparedness, data privacy, ethical concerns, and equity of access. The paper concluded that AI-driven flipped classrooms could transform education if implemented responsibly with strong institutional support and inclusive policies.

Martin, F. [10] reviewed the role of e-learning and information technology in enhancing teaching and learning in higher education. The paper explored how learning management systems and online tools support communication, collaboration, and access to educational resources. It highlighted both the pedagogical benefits of e-learning and the need for changes in instructional design. It also addressed challenges such as faculty readiness, technological infrastructure, and student support. The study concluded that effective integration of e-learning required strategic planning, training, and institutional commitment.

Mahmadoov, Y. [11] reviewed the importance of Learning Management Systems (LMS) in modern education and how they supported online and distance learning. The paper explained that LMS platforms helped manage course materials, communication, assignments, assessments, and student progress in a centralized system. The study highlighted the advantages of LMS such as flexibility, improved access to learning resources, and better interaction between teachers and students. It also described common LMS features including course management, content delivery, tracking of learning activities,

and evaluation tools. Technology concluded that LMS technology played a vital role in improving the quality and efficiency of education in the digital learning environment.

Phakhot, N., & Attahakul, P [12] reviewed how Learning Management Systems (LMS) were revolutionizing modern education by integrating digital technologies into the learning process. They explained that LMS platforms supported online course delivery, communication, assessment, and management of learning resources. The study highlighted benefits such as improved accessibility, flexibility, and enhanced student engagement in the learning environment. It also noted challenges including technological barriers, lack of digital skills, and infrastructure limitations in some institutions. Overall, the paper concluded that LMS played a vital role in transforming traditional education into a more efficient and technology-driven system.

Kokkinos, L. [13] reviewed how Information and Communication Technology (ICT) had transformed modern education by integrating digital tools into teaching and learning. The paper discussed the evolution of educational technology, including computers, the internet, interactive whiteboards, mobile devices, and online learning platforms. These technologies made learning more interactive, personalized, and accessible for students. It also highlighted the rapid adoption of remote learning tools during the COVID-19 pandemic. Overall, technology played a crucial role in creating flexible, collaborative, and future-ready educational environments.

Zhang, Z. [14] reviewed how educational technology was transforming online education through digital learning platforms, personalized learning systems, and immersive technologies like virtual and augmented reality. These technologies improved accessibility, flexibility, and student engagement in online learning environments. The paper also highlighted benefits such as interactive learning, collaboration, and data-driven decision making for educators. However, it discussed challenges including technological limitations, the need for pedagogical adaptation, and privacy or security

concerns. Overall, the study concluded that educational technology played a crucial role in shaping the future of modern online education.

Cabrera, B. C. C. [15] reviewed research trends on the integration of Artificial Intelligence (AI) in Learning Management Systems (LMS) using a bibliometric analysis method. The study examined 261 research articles from databases such as Scopus and Web of Science to identify key authors, journals, keywords, and publication trends. Results showed a rapid increase in research on AI-based LMS, especially around 2015–2021, highlighting growing interest in personalized and adaptive learning. Major themes included e-learning, machine learning, learning styles, and AI tools that improved content personalization and student performance. The paper concluded that although AI enhanced LMS capabilities, more research was needed on effectiveness, user acceptance, and ethical issues such as privacy and bias.

2.2 Analysis of gaps or limitations in current solutions

Many existing Revolutionizing Digital Learning (RDL) Platforms are designed for general educational purposes and are not customized for specific departments. Because of this, they often fail to effectively support structured academic programs such as the IT Department curriculum, which requires organized semester-wise, course materials, programming resources.

Popular platforms such as Coursera, Udemy and YouTube provide many courses, but they are not aligned with the specific syllabus of the University of the Punjab BS.IT degree Program. Most of these platforms provide courses in English and many of them are paid, which creates accessibility issues for students. Furthermore, YouTube playlists do not cover all subjects of a semester in a single organized structure according to the official course outline.

Another limitation is that there is no single platform that provide structure semester-wise content, and assignments specifically for the University of the Punjab BS.IT students. Students often need to search multiple platforms to find study materials, which makes the learning process inefficient and time consuming.

2.3 Justification for your approach

To overcome these limitations, our project introduces a specialized Revolutionizing Digital Learning With AI (RDL) system that will be designed specifically for BS-IT students at the University of the Punjab. The platform will organize course content according to the official semester-wise curriculum, making it easier for students to access relevant study materials.

The system will integrate multiple academic resources in one place, including YouTube Video lectures, and assignment. This will reduce students' dependency on multiple platforms and provide a centralized learning environment. In addition, our system will include an AI Chatbot to response students queries and provide quick support during their studies. The user interface will be simple, lightweight, and easy to navigate, ensuring that students can efficiently access the required content without unnecessary complexity.

Moreover, our platform will provide free access to course materials specifically for the University of the Punjab BS.IT degree program. By focusing on the exact needs of the students and aligning the content with their syllabus, our system aims to improve learning efficiency of the students.

CHAPTER 3

SYSTEM ANALYSIS

3.1 Requirements Gathering

Requirements Gathering is the process of collecting, analyzing, and documenting the needs and expectations of stakeholders to define what the system should be to do. For Revolutionizing Digital Learning With AI, this ensures the platform meets the needs of Students, Teachers, and Admins.

3.2 Functional Requirements

Table 3.1: FR-1 Signup

Identifier	FR-1
Title	Signup
Description	The System allows users to Sign up with Name, email, Password, & Role
Rationale	Required to create a unique password for each user
Dependencies	Internet Connection
Error Handling	If email already exists, show error message.

Table 3.2: FR-2 Login

Identifier	FR-2
Title	Login
Description	The System allows users to Login with email, & Password
Rationale	Provide secure access to the system
Dependencies	Internet Connection, FR-1
Error Handling	If incorrect credentials enter, show error message.

Table 3.3: FR-3 Google Authentication

Identifier	FR-3
Title	Google Authentication
Description	The System allows users to Sign up or Login via their google account in a single step
Rationale	Required to create a unique password for each user
Dependencies	Internet connection, Firebase Authentication
Error Handling	If don't have google account, show error message.

Table 3.4: FR-4 Reset Password

Identifier	FR-4
Title	Reset Password
Description	The System allows users to reset password through OTP.
Rationale	Helps user recover account in case of lost password.
Dependencies	Internet connection, FR-2
Error Handling	If entered email does not exist, system should display "Email not exist".

Table 3.5: FR-5: Edit Profile

Identifier	FR-5
Title	Edit Profile
Description	The System allows users to update personal information like avatar (photo), name, description.
Rationale	Allow user to see their account and update it
Dependencies	Internet connection, FR-2
Error Handling	If invalid file format for avatar, show error message.

Table 3.6: FR-6 View Courses

Identifier	FR-6
Title	View Courses
Description	The System shall display list of courses to users from Database.
Rationale	Provide learning resources to students
Dependencies	Internet connection, FR-11
Error Handling	If no courses are available, display “No courses found yet”.

Table 3.7: FR-7 View Lectures

Identifier	FR-7
Title	View Lectures
Description	The System allows users to view lecture including YouTube videos along with content.
Rationale	Provide detailed learning material to students
Dependencies	Internet connection, FR-6
Error Handling	If no lectures are available, display “No lectures found yet”.

Table 3.8: FR-8: View Assignments

Identifier	FR-8
Title	View Assignments
Description	The System allows users to view Assignments task, live timer and perform it.
Rationale	Test user's as general knowledge
Dependencies	Internet connection, FR-6
Error Handling	If no assignment are available, display "No assignment found yet".

Table 3.9: FR-9: Submit Assignment

Identifier	FR-9
Title	Submit Assignment
Description	The System allows users to perform assignment task and submit before timer expired.
Rationale	Test user's as general knowledge
Dependencies	Internet connection, FR-8
Error Handling	If timer end, display "timer has expired".

Table 3.10: FR-10: AI Chatbot

Identifier	FR-10
Title	Ai chatbot
Description	The System allows user to help Ai chatbot for lectures.
Rationale	Test user's as general knowledge
Dependencies	Internet connection, FR-6
Error Handling	If timer end, display "timer has expired".

Table 3.11: FR-11: Create Course

Identifier	FR-11
Title	Create Course
Description	The System allows teachers to add, update & delete course.
Rationale	Ensure courses are up to date.
Dependencies	Internet connection, FR-1
Error Handling	If database not connected, show error message.

Table 3.12: FR-12: Create Lecture

Identifier	FR-12
Title	Create Lecture
Description	The System allows teachers to add lectures by YouTube link along with content. And also update & delete this.
Rationale	Provide detailed learning resources
Dependencies	Internet connection, FR-11
Error Handling	If database not connected, show error message.

Table 3.13: FR-13: Create Assignment

Identifier	FR-13
Title	Create Assignment
Description	The System allows teachers to add assignment tasks along with set timer. And update & delete this.
Rationale	Keeps assignment relevant to learning content
Dependencies	Internet connection, FR-11
Error Handling	If database not connected, show error message.

Table 3.14: FR-14: Manage Course

Identifier	FR-14
Title	Manage Course
Description	The System allows admin to manage courses (Add, update & delete)
Rationale	Ensure courses are up to date.
Dependencies	Internet connection, FR-1
Error Handling	If database not connected, show error message.

Table 3.15: FR-15 Manage User

Identifier	FR-14
Title	Manage User
Description	The System allows admin to manage users (Add & delete)
Rationale	Ensure users are registered.
Dependencies	Internet connection, FR-1
Error Handling	If database not connected, show error message.

Table 3.16: Non-functional Requirements

ID	Non-Functional Requirement	Details
NFR-1	Performance	The system shall load courses, lectures and assignments within 2-3 seconds under normal internet connection.
NFR-2	Security	The system shall store user passwords in encrypted form and allow only register user to log in.
NFR-3	Usability	The system shall provide a simple and user-friendly interface with sidebar navigation and clear layouts.
NFR-4	Reliability	The system shall maintain ensure data accuracy and prevent data loss
NFR-5	Availability	The system shall remain available to users 24/7, via internet.
NFR-6	Scalability	The system shall support multiple users accessing the system at the same time without performance issues.
NFR-7	Maintainability	The system shall allow admin to manage courses and user easily through database without coding
NFR-8	Compatibility	The system shall work properly on different devices such as laptops, tablets, and mobiles.

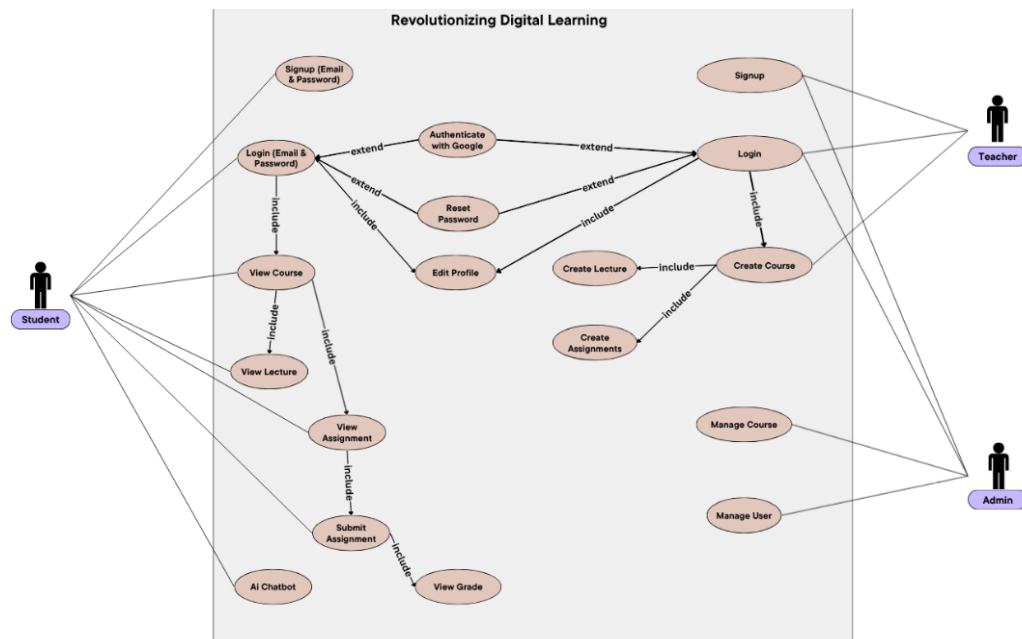


Figure 3.1: Use Case Diagrams / Descriptions

Table 3.17: UC-1: Signup

Identifier	UC-1
Title	Email-Password Signup
Actor	User (Student, Teacher & admin)
Preconditions	User is on Signup page. Internet connection is available.
Postconditions	New user account created and redirected to home page.
Normal Flow	1. User enters Name, email, password, and role 2. User Click Signup . 3. System validates credentials and store account in Database. 4. User is signup successfully and redirected to home page.
Alternative Flow	A1: Password don't match → Error Shown A2: Email already exists → Error Shown A3: Internet not available → Error Shown

Table 3.18: UC-2: Login

Identifier	UC-2
Title	Email-Password Login
Actor	User (Student, Teacher & admin)
Preconditions	User is on Login page. Internet connection is available.
Postconditions	User log in and redirected to home page.
Normal Flow	1. User enters email, & password. 2. User Click Login . 3. System validates credentials 4. User is Login successfully and redirected to home page.
Alternative Flow	A1: Invalid credentials → Error Shown A2: Internet not available → Error Shown

Table 3.19: UC-3: Google Authentication

Identifier	UC-3
Title	Google Authentication
Actor	User (Student, Teacher & admin)
Preconditions	User is on Signup or Login page. Internet connection is available.
Postconditions	User signup or log in by single click and redirected to home page.
Normal Flow	1. User select “Continue with Google”. 2. Google signup or login dialog appears. 3. User select Google account . 4. System validates credentials 5. User is Signup or Login successfully and redirected to home page.
Alternative Flow	A1: Internet not available → Error Shown

Table 3.20: UC-4: Reset Password

Identifier	UC-4
Title	Reset Password
Actor	User (Student, Teacher & admin)
Preconditions	User is on Login page. Internet connection is available.
Postconditions	Password reset OTP sent to user's email
Normal Flow	1. User click "Forget Password". 2. User enter verify email 3. System send OTP to email 4. User reset password using OTP.
Alternative Flow	A1: Email not found → Error Shown A2: Internet not available → Error Shown

Table 3.21: UC-5: Update Profile

Identifier	UC-5
Title	Update Profile
Actor	User (Student, Teacher & admin)
Preconditions	User is log in
Postconditions	Update profile save din database
Normal Flow	1. User Click circle from navbar and Click My Profile and Click on Edit Profile 2. User edit avatar(photo) ,name, & description 3. User Click Save 4. System update profile in database
Alternative Flow	A1: Internet not available → Error Shown

Table 3.22: UC-6: Display Course list

Identifier	UC-6
Title	Display Course list
Actor	User (Student, Teacher & admin)
Preconditions	1. User is on Home or Dashboard Page → Click “All Courses”
Postconditions	System Show list of available courses from Database
Normal Flow	1. User Click Courses . 2. System fetches course list from Database 3. User view list of all available courses.
Alternative Flow	A1: No course available → Display “No Courses found yet” A2: Internet not available → Error Shown

Table 3.23: UC-7: View Course

Identifier	UC-7
Title	View Course
Actor	User (Student, Teacher & admin)
Preconditions	User clicked on any course
Postconditions	User navigate view course page
Normal Flow	1. User Click any Course . 2. System navigate view course page 3. User view the detail of course
Alternative Flow	A1: No course available → Display “No Courses found yet” A2: Internet not available → Error Shown

Table 3.24: UC-8: View Lecture

Identifier	UC-8
Title	View Lecture
Actor	User (Student, Teacher & admin)
Preconditions	User clicked on Enroll
Postconditions	User navigate view lecture page
Normal Flow	1. User Click Enroll & watch. 2. System navigate to View lecture page 3. User can watch video and read content from left side. 4. User view and select list of available lecture
Alternative Flow	A1: No lecture available → Display “No lecture found yet” A2: Internet not available → Error Shown

Table 3.25: UC-9: View Assignment

Identifier	UC-9
Title	View Assignment
Actor	User (Student, Teacher & admin)
Preconditions	User clicked on Assignment
Postconditions	User navigate view assignment page
Normal Flow	1. User Click Assignment. 4. System navigate to View Assignment page 5. User can view assignment task, perform it, and submit before timer expired.
Alternative Flow	A1: No assignment available → Display “No assignment found yet” A2: Internet not available → Error Shown

Table 3.26: UC-10: Submit Assignment

Identifier	UC-10
Title	Submit Assignment
Actor	User (Student, Teacher & admin)
Preconditions	User perform Assignment task
Postconditions	User submit assignment
Normal Flow	User can perform assignment task submit before timer expired.
Alternative Flow	A1: Timer expired → Display “timer expired” A2: Internet not available → Error Shown

Table 3.27: UC-11: Create Course

Identifier	UC-11
Title	Create Course
Actor	Teacher
Preconditions	Teacher must sign up or login
Postconditions	Course Create in database and available for students.
Normal Flow	1. Teacher Click Course from Sidebar. 2. System navigate Create course page 3. Enter course title and click on save. 4. Course are created. 5. System update database
Alternative Flow	A1: Internet not available → Error Shown

Table 3.28: UC-12: Create Lecture

Identifier	UC-12
Title	Create Lecture
Actor	Teacher
Preconditions	User clicked on Lecture
Postconditions	User navigate create lecture page
Normal Flow	1. User Click “go to lecture” from edit course page. 2. System navigate to create lecture page 3. Enter title, save and navigate to edit lecture page. 4. Enter YouTube video link and content 5. Click save
Alternative Flow	A1: Internet not available → Error Shown

Table 3.29: UC-13: Create Assignment

Identifier	UC-13
Title	Create Assignment
Actor	Teacher
Preconditions	User clicked on Assignment
Postconditions	User navigate create assignment page
Normal Flow	1. User Click “go to assignment” from edit course page. 2. System navigate to create assignment page 3. Enter title, save and navigate to edit assignment page. 4. Enter content and set timer 5. Click save
Alternative Flow	A1: Internet not available → Error Shown

Table 3.30: UC-14: Manage Course

Identifier	UC-14
Title	Manage Course
Actor	Admin
Preconditions	Admin must be signup or login
Postconditions	User navigate create lecture page
Normal Flow	1. User Click “go to lecture” from edit course page. 2. System navigate to create lecture page 3. Enter title, save and navigate to edit lecture page. 4. Enter YouTube video link and content 5. Click save
Alternative Flow	A1: Internet not available → Error Shown

Table 3.31: UC-15: Manage User

Identifier	UC-15
Title	Manage user
Actor	Admin
Preconditions	User clicked on Lecture
Postconditions	User navigate create lecture page
Normal Flow	1. User Click “go to lecture” from edit course page. 2. System navigate to create lecture page 3. Enter title, save and navigate to edit lecture page. 4. Enter YouTube video link and content 5. Click save
Alternative Flow	A1: Internet not available → Error Shown

CHAPTER 4

SYSTEM DESIGN

4.1 System Architecture

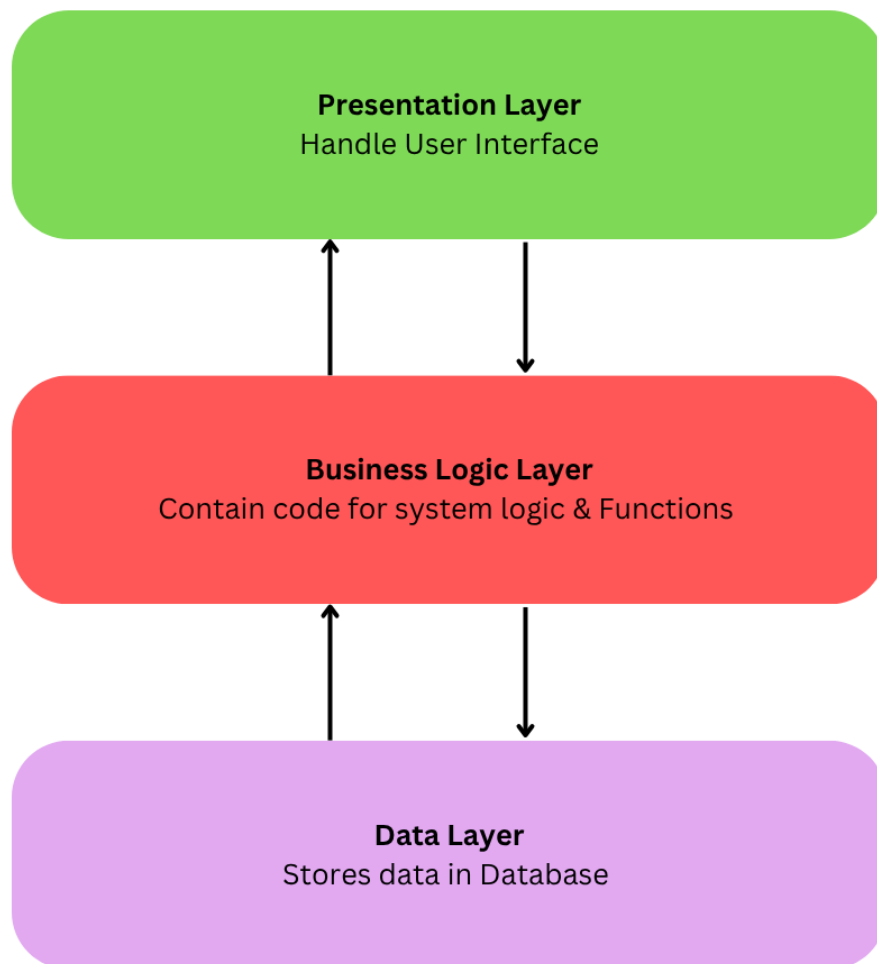


Figure 4.1: System Architecture

4.2 Database Design

Database Design is the process of creating the structure of a database, including tables, fields, relationships, and keys to store system data efficiently.

For Revolutionizing Digital Learning With AI Project, here is a professional database design suitable for MERN Stack (MongoDB) or relational databases.

Main Entities

1. Users
2. Courses
3. Lectures
4. Assignments
5. Submissions
6. AI Chatbot

4.2.1 Database Table Design

Table 4.1: User Table

Field Name	Data Type	Description
user_id	ObjectId / INT	Primary Key
name	String	User full name
email	String	User email
password	String	Encrypted password
role	String	Admin / Teacher / Student

Table 4.2: Course Table

Field Name	Data Type	Description
course_id	ObjectId / INT	Primary Key
course_title	String	Course name
description	String	Course details
teacher_id	String	Foreign key from Users

Table 4.3: Lecture Table

Field Name	Data Type	Description
lecture_id	ObjectId / INT	Primary Key
course_id	ObjectId	Foreign key from Courses
lecture_title	String	Lecture name
video_url	String	Lecture video link
content	String	Lecture Content

Table 4.4: Assignment Table

Field Name	Data Type	Description
assignment_id	ObjectId	Primary Key
course_id	ObjectId	Foreign key
title	String	Assignment title
description	String	Assignment details
due_date	DATETIME	Submission deadline

Table 4.5: Submission Table

Field Name	Data Type	Description
submission_id	ObjectId	Primary Key
assignment_id	ObjectId	Foreign key
student_id	ObjectId	Foreign key
file_url	String	Submitted file
submitted_at	DATETIME	Submission date

One Teacher → Many Courses

One Course → Many Lectures

One Student → Many Enrollments

One Course → Many Assignments

One Assignment → Many Submissions

4.3 Interface Design

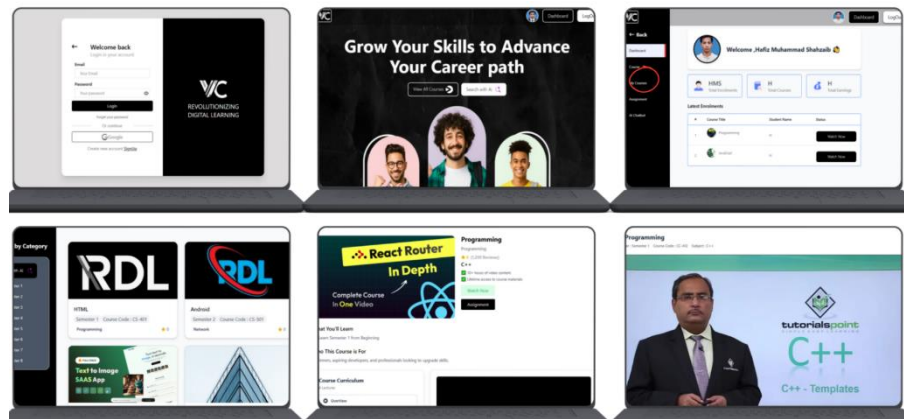


Figure 4.2: Screen images

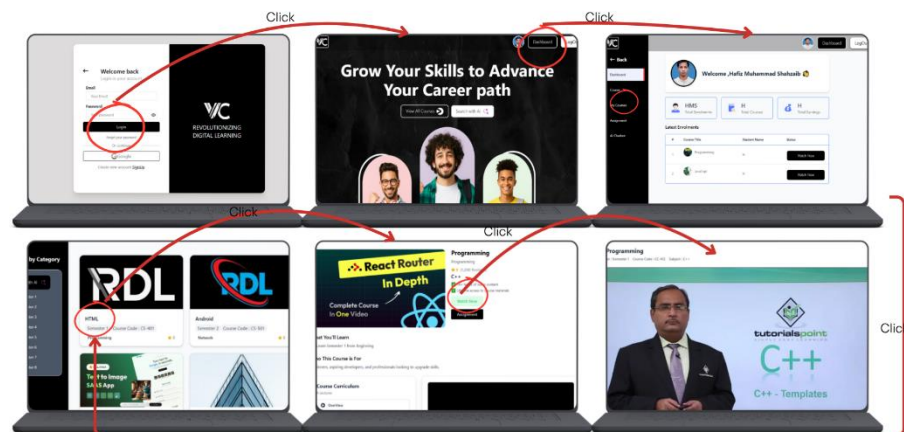


Figure 4.3: Screen Objects & Actions:

4.4 UML Diagrams (Class Diagrams, Sequence Diagrams, etc.)

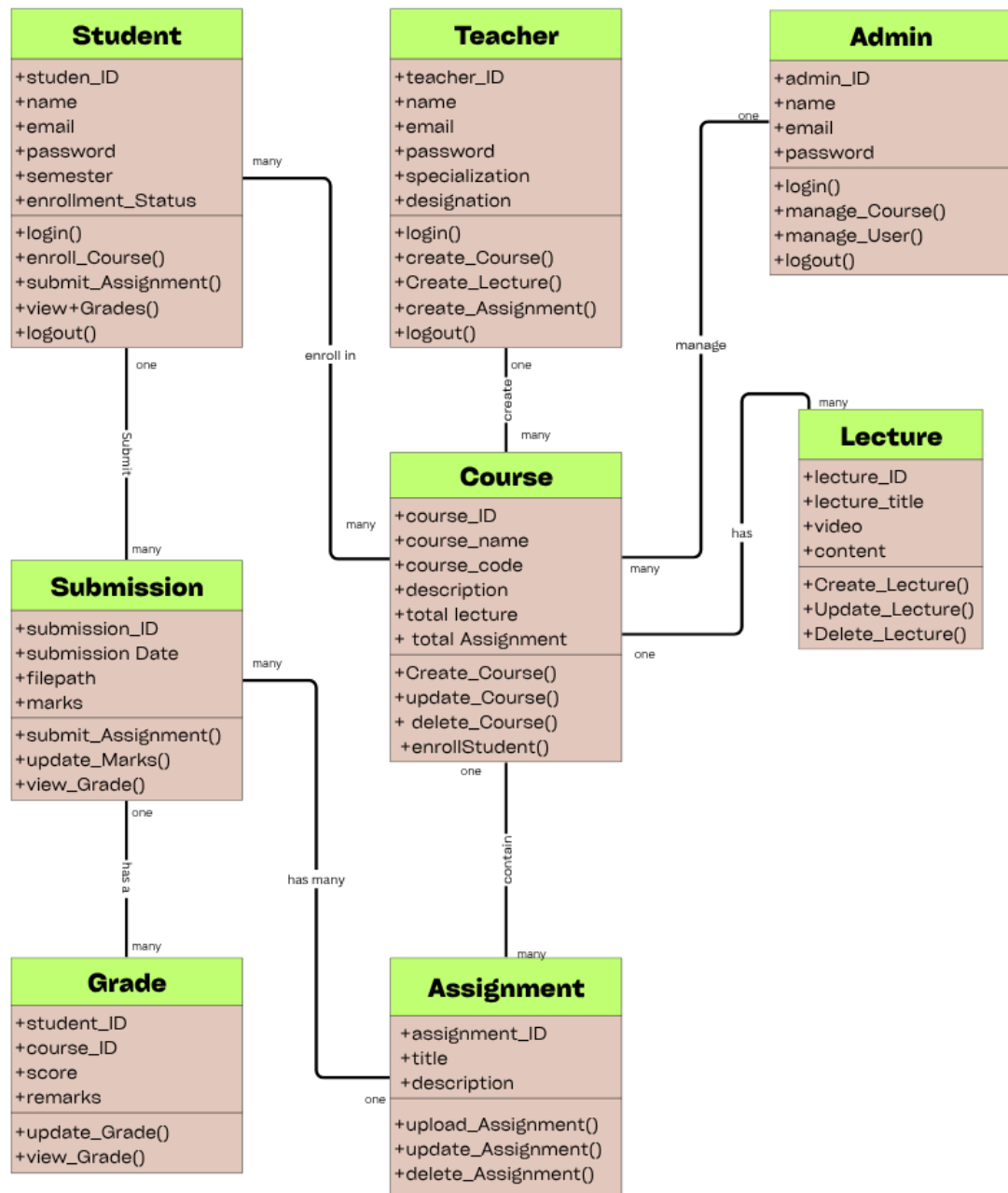


Figure 4.4: Class Diagrams

4.4 Sequence diagrams

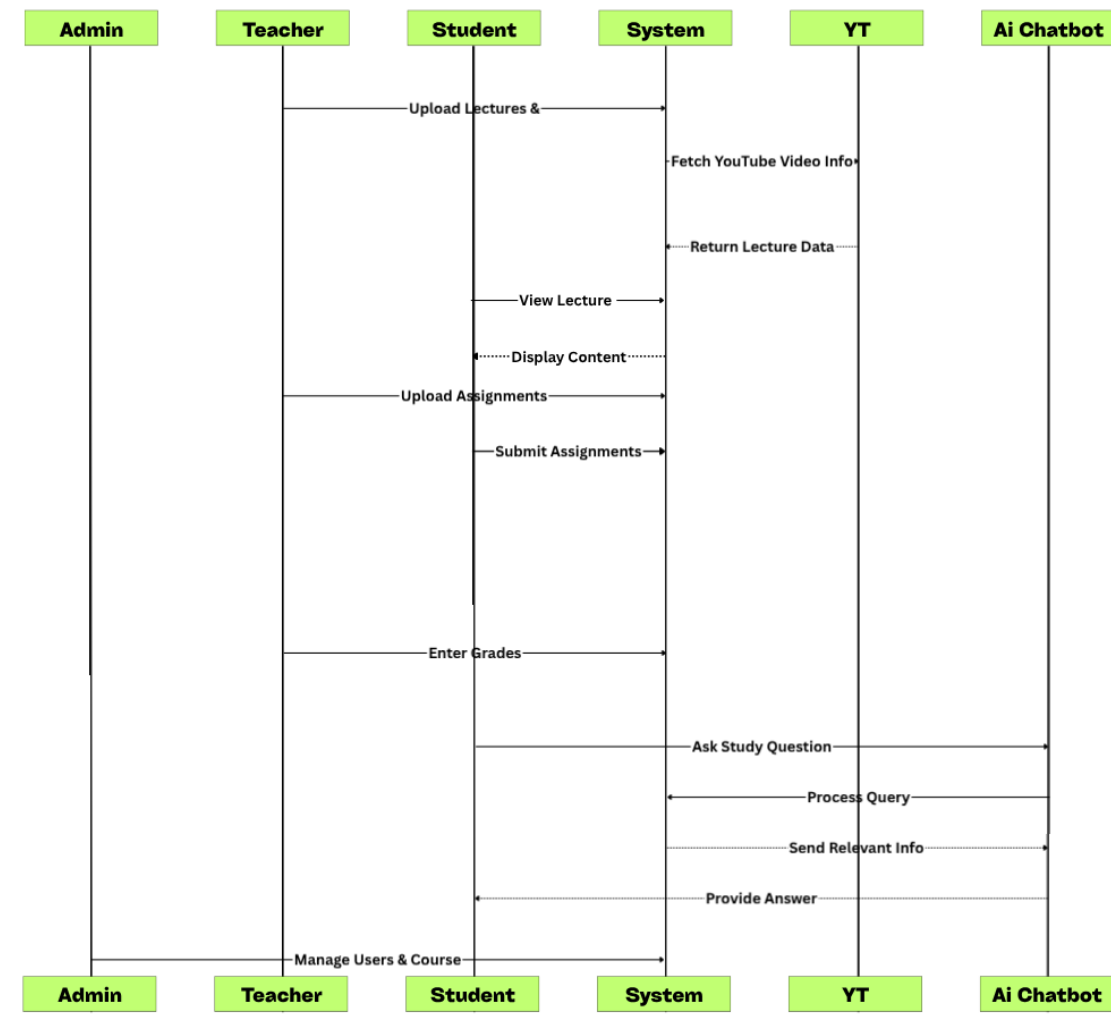


Figure 4.5: Class Diagrams

CHAPTER 5

IMPLEMENTATION

5.1 Development Tools and Environment

Table 5.1: Development Tools

Category	Examples of Tools
Programming Language	JavaScript
Web Frameworks	React, Express(Node.js)
DataBase	MongoDB
State Management	Redux ToolkitRedux Toolkit
API Integration	Axios
AI Integration	Gemini API
Design	Canva, Figma
Cybersecurity Tools	JWT Authentication, Password Hashing(bcrypt)
Image Handling	Image Kit / Cloudinary
API Testing	Postman ()
Version Control	Git / GitHub
Package Manager	npm

Table 5.2: Development Environment

Category	Environment
Operating System	Window 10
Code Editor	VS Code
Browser	Google Chrome
Server Environment	Local Host
Hardware	RAM (4GB), Hard Disk(500GB)

5.2 Modules Implemented

When a user visits the website, the Home Screen checks the user data. If the user is authenticated, the navbar displays the Dashboard and Logout buttons; otherwise, it shows only the Login button.

5.2.1 Module - 1: User Authentication

1. Signup with name, email, password & role (or Google)
2. Login with email & password (or Google)
3. Reset password option.
4. Password visibility toggle (eye icon)

5.2.2 Module - 2: Courses

1. Display list of courses
2. Select course -> navigate to view course -> Enroll & watch

5.2.3 Module - 3: Lectures

1. From view course -> Enroll & watch -> Display list of lecture
2. Select lecture -> Display video with play icon & content.

5.2.4 Module - 4: Assignment:

1. From view course -> Assignment

2. Display content, live timer & submit button.

3. After submission grade

5.2.5 Module - 5: Profile

1. View and edit profile details.

2. Update avatar, name & description

5.2.6 Module - 6: Create Course:

1. Teacher add new course with title & description

2. Update existing course

3. Delete course

5.2.7 Module - 7: Create Lecture:

1. Teacher add new lecture with YT video link & content

2. Update existing lecture

3. Delete lecture.

5.2.8 Module - 8: Create Assignment:

1. Teacher add new Assignment with content & ending date

2. Update existing Assignment or timer

3. Delete Assignment.

5.2.9 Module - 9: Manage Course:

1. Admin add new course

2. Update existing course

3. Delete course

5.2.10 Module - 10: Manage User:

1. Admin add new user with email, name, password & role

2. Delete user

5.2.11 Module - 11: Ai Chatbot

1. Student generate response by given input.
2. Understand and overcome lectures issues.

CHAPTER 6

TESTING

6.1 Testing Techniques Used

To ensure that Revolutionizing Digital Learning functions correctly and meets the defined requirements, a comprehensive testing strategy was adopted. Two primary testing techniques were applied: Black Box testing and White Box testing. These techniques complement each other by verifying external behavior as well as internal logic, providing full coverage of the system's functionality.

6.1.1 Black Box Testing

Black Box testing focuses on verifying the system outputs based on given inputs without any knowledge of internal code structure. In Revolutionizing Digital Learning, this technique was used to test all modules such as user authentication, courses, lecture, assignment and AI Chatbot. Test cases were designed based on the functional requirements defined in Chapter 3. Testers interacted with the system through the web interface and verified that each function produced the correct and expected results. This approach ensure that the system behaves correctly from the end-user's perspective.

6.1.2 White Box Testing

White Box Testing examines the internal logic and structure of code. In Revolutionizing Digital Learning, this technique was applied to all the modules of the system. Unit tests were written to all the individual modules of the system. This approach ensures that the internal processing of data is working properly.

6.2 Test Cases / Test Plans

Table 6.1: Test Case

Test Case	Title	Precondition	Test Step	Input	Expected Result	Actual Result	Status
TC-1	Email-Password Signup	Database is connected	1.open web page → Login → Signup 2. Enter Valid data 3. Click Signup	Name, Email , Passswor d, & Role	User Signup and redirecte d to home page	Signup Successfull y	Pass
TC-2	Email-Password Signup Error	Database is connected	1.open web page → Login → Signup 2. Enter InValid data 3. Click Signup	Name, Email , Passswor d, & Role	Error: Email already exist	same	Pass
TC-3	Email-Password Login	User exist in DB	1.open web page → Login 2. Enter email & Passwor d 3. Click Login	Email & Password	User Login and redirecte d to home page	Login Successfull y	Pass
TC-4	Email-Password Login Error	User exist in DB	1.open web page → Login 2. Enter invalid data 3. Click Login	Email & Password	Error: Email doesn't exist	same	Pass
TC-5	Forget Password	User exist in DB	1.open web page → Login 2. Click "Forget Passwor d" 3. Enter valid	Email	Passwor d OTP sent	Email sent successfull y	Pass

			email				
TC-6	Google Authentication	Google account and DB connected	1.open web page → Login 2. Click “Continue with Google”	NA	User signup or login & redirected to home page	Same	Pass
TC-7	View Profile	User exist in DB	1.open web page → Circle→ My Profile 2. Click Edit 3. Change data except email & role	NA	Update Profile	Same	Pass

Table 6.2: For Student Side

TC-8	Courses	Course exist in DB	1. open sidebar 2. Click Course	NA	Course list displayed	Same	Pass
TC-9	View lecture	Course & Lecture exist in DB	1.select any course 2. Click “Enroll & Watch”	NA	Video and content displayed	Same	Pass
TC-10	View Assignment	Course & Assignment exist	1.select any course 2. Click “Assignment”	NA	Assignment displayed	Same	Pass
TC-11	View Grade	Submit Assignment	1.select any course 2. Click “Assignment” 3. Click Submit	NA	Grade Displayed	Same	Pass
TC-12	AI Chatbot	DB is connected	1.open sidebar 2. Click “Ai Chatbot”	NA	Generate response on the basic of user queries	Same	Pas

Table 6.3: For Teacher Side

TC-13	Create Course	DB is connected	1.open Sidebar 2. click course 3. click create course	Title, Description	Course Created and save in DB	Same	Pass
TC-14	Create Lecture	Course exist in DB	1.open side bar 2. click course 3. click edit icon 4. Click “go to lecture page”	Title, Content, YouTube link	Lecture Created & save in DB	Same	Pass
TC-15	Create Assignment	Course & lecture exist in DB	1.open side bar 2. click course 3. click edit icon 4. Click “go to assignment page”	Title, content, set Timer	Assignment Created & save in DB	same	Pass

Table 6.4: For Admin Side

TC-16	Manage Course	DB is connected	1.open sidebar 2. click Add course	NA	Handle all course data	same	Pass
TC-17	Manage User	DB is connected	1.open sidebar 2. click Add user	NA	Handle all user	Same	Pass

Table 6.5: Results and Bug Reports

	Total Test	Passed	Failed	Success Rate
User	7	7	0	100%
Student	5	5	0	100%
Teacher	3	3	0	100%
Admin	2	2	0	100%
Overall Result	17	17	0	100%

6.2.1 Bugs Reports:

During development and testing, several bugs and system issues were identified. These issues were analyzed and fixed to improve system performance, usability, and security.

Table 6.6: Bugs reports

Bug ID	Module	Problem Identified	Cause	Solution Applied	Status
BR-1	Authentication	User session expired after page refresh	JWT token was not stored properly	Implemented token storage and authentication middleware	Fixed
BR-2	Login System	Invalid password error not handled properly	Missing validation handling	Added backend validation and error messages	Fixed
BR-3	Course Module	Courses were not loading instantly	Slow API response	Optimized API calls using Axios and Redux Toolkit	Fixed
BR-4	Lecture Module	Video upload caused slow loading	Large video file upload issue	Replaced direct upload with YouTube URL integration	Fixed
BR-5	Assignment Module	Assignment timer did not update correctly	Timer state synchronization issue	Implemented real-time timer logic using React hooks	Fixed
BR-6	AI Chatbot	AI response delay for long queries	Slow API processing	Improved request handling and optimized API integration	Fixed

BR-7	Profile Module	Avatar image not updating immediately	Cache issue after upload	Added image refresh handling with Cloudinary/ImageKit	Fixed
BR-8	Dashboard	Unauthorized dashboard access	Missing role-based route protection	Implemented protected routes and role verification	Fixed

CHAPTER 7

RESULTS AND DISCUSSION

Table 7.1 Performance Metrics

#	Title	Description	Expected Result	Actual Result	Status
P-1	Authentication Time	User verification	2-3 seconds	2 sec	Pass
P-2	Course Loading Time	Load available course	<5 sec	3 sec	Pass
P-3	Lecture Loading Time	Load available lecture	within 4 second.	3.5 sec	Pass
P-4	Assignment Loading Time	Load available assignment	within 4 second.	3.5 sec	Pass
P-5	Assignment Submission Time	Submission send time	within 2 second	2 sec	Pass
P-6	Ai Response Time	Generate Response Time	Within 5 second	4.5 sec	Pass

Table 7.2: Teacher Side

P-7	Create Course Time	Course send time	within <3 sec	2.5 sec	Pass
P-8	Create Lecture Time	Lecture send time	within <4 second	3.8 sec	Pass
P-9	Create Assignment Time	assignment send time	within <4 second	3.8 sec	Pass

Table 7.3: Admin Side

P-10	Add Course Time	Course send time	Add Course in <3 second	2.5 sec	Pass
P-11	Add User Time	User send time	Add User in <4 second	3.8 sec	Pass

7.1 Evaluation of Results

The system performance was evaluated using response and loading time measurement.

Table 7.4: Student Side:

Feature	Expected Performance	Actual Performance	Evaluation
Authentication	2–3 seconds	2 seconds	Efficient
Course Loading	<5 seconds	3 seconds	Good
Lecture Loading	<4 seconds	3.5 seconds	Good
Assignment Loading	<4 seconds	3.5 seconds	Good
Assignment Submission	<2 seconds	2 seconds	Efficient
AI Chatbot Response	<5 seconds	4.5 seconds	Acceptable

Table 7.5: Teacher Side

Course Creation	<3 seconds	2.5 seconds	Efficient
Lecture Creation	<4 seconds	3.8 seconds	Good
Assignment Creation	<4 seconds	3.8 seconds	Good

Table 7.6: Admin Side

Add Course	<3 seconds	2.5 seconds	Efficient
Add User	2–3 seconds	2 seconds	Efficient

CHAPTER 8

CONCLUSION & FUTURE WORK

Table 8.1: Summary of Achievements

Objective	Status
Implement Secure Email and Google based authentication	Completed
Integrate User role-based & Dashboard access (Student, Teacher & Admin)	Completed
Responsive design for large & Small Screen	Completed
Allow Course Enrollments	Completed
Allow take Lecture	Completed
Allow AI Chatbot help	Completed
Allow take Assignment	Completed
Allow User to perform Assignments tasks & View Grades	Completed
Allow Course Rating	Completed
Allow Create Course	Completed
Allow Create Lecture	Completed
Allow Create Assignments	Completed
Manage Course	Completed
Manage User	Completed

8.1 Challenges Faced

During development of our project, several challenges were encountered.

Challenge: Again, log in & Signup to revisit the website.

Solution: To overcome this issue, JWT (JSON Web Token) based authentication implemented in the backend

Challenge: To upload video directly which cause Slow upload and loading time for lectures

Solution: To overcome this issue, the video upload system was replaced with YouTube video URL integration.

8.2 Suggestions for Further Improvement

The current version focuses on main features, but future improvements can make it more advanced:

- AI Quiz Generator
- Android & IOS Version
- Offline access to courses, lecture & assignments for areas with weak internet,

And a powerful backend server and many features will be added.

APPENDICES

GitHub Link

<https://github.com/Hafiz-Shahzaib/FYP.git>

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